

Introduction to Loops

Understanding Iteration using Flowcharts

Programming Fundamentals Team

SETU

2026-06-17

Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
4. Infinite loops
5. Useful loop patterns
6. Activities

Outline

1. Why do we need loops?

2. Selection versus Iteration

3. Anatomy of a loop

4. Infinite loops

5. Useful loop patterns

6. Activities

1.1 What if we needed to repeat this?

[Without loops, repeated tasks quickly become messy:

```
print("Welcome")  
print("Welcome")  
print("Welcome")  
print("Welcome")
```

A loop lets us write the repeated instruction once.],

1.1 What if we needed to repeat this?

Key idea

Computers are very good at repetition.

1.2 Everyday repetition

- Checking each student's mark
- Taking orders until the queue is empty
- Asking for a password until it is correct
- Displaying a menu until the user chooses Exit
- Reading temperatures from a sensor

Loops are used when a set of instructions must be repeated.

1.3 What is iteration?

Iteration means repeating a set of instructions until a condition is met.

Examples:

- Continue asking until the input is valid.
- Continue counting until the target is reached.
- Continue processing until there is no more data.

Outline

1. Why do we need loops?

2. Selection versus Iteration

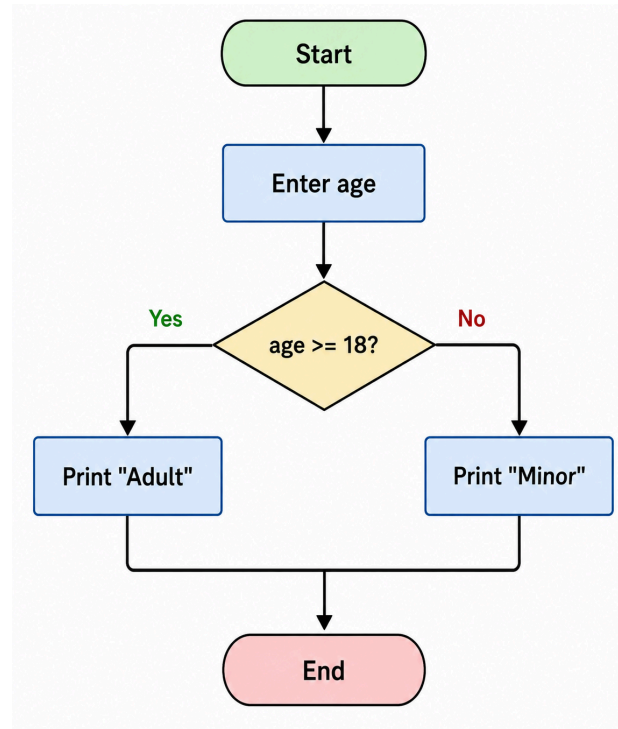
3. Anatomy of a loop

4. Infinite loops

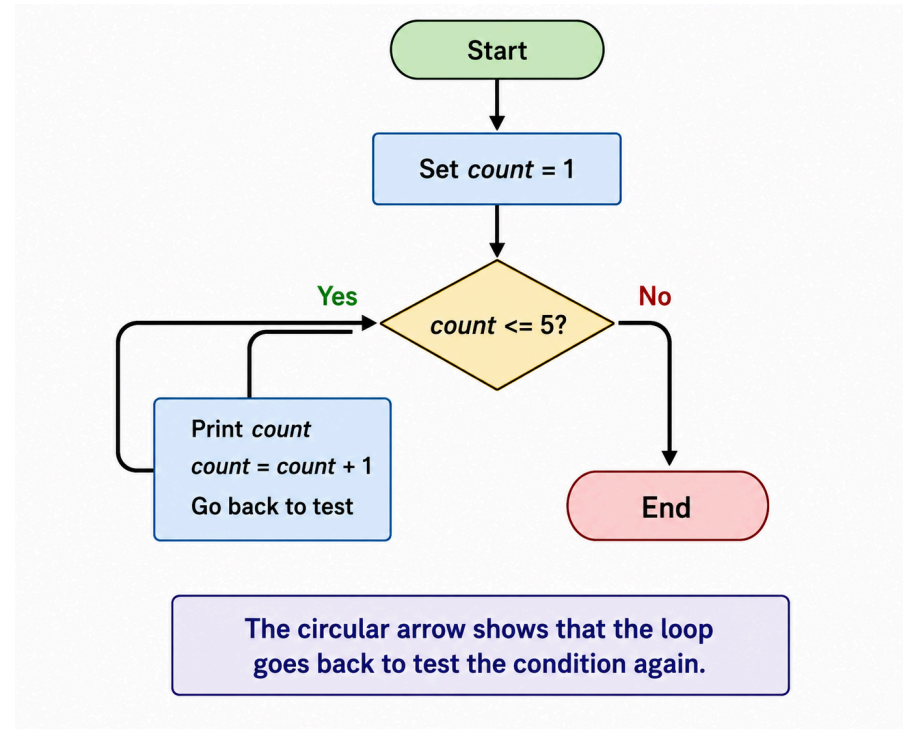
5. Useful loop patterns

6. Activities

2.1 Selection asks once



2.2 Iteration asks repeatedly



2.3 The key difference

Selection

- Condition checked once
- One path is chosen
- Program continues

2.3 The key difference

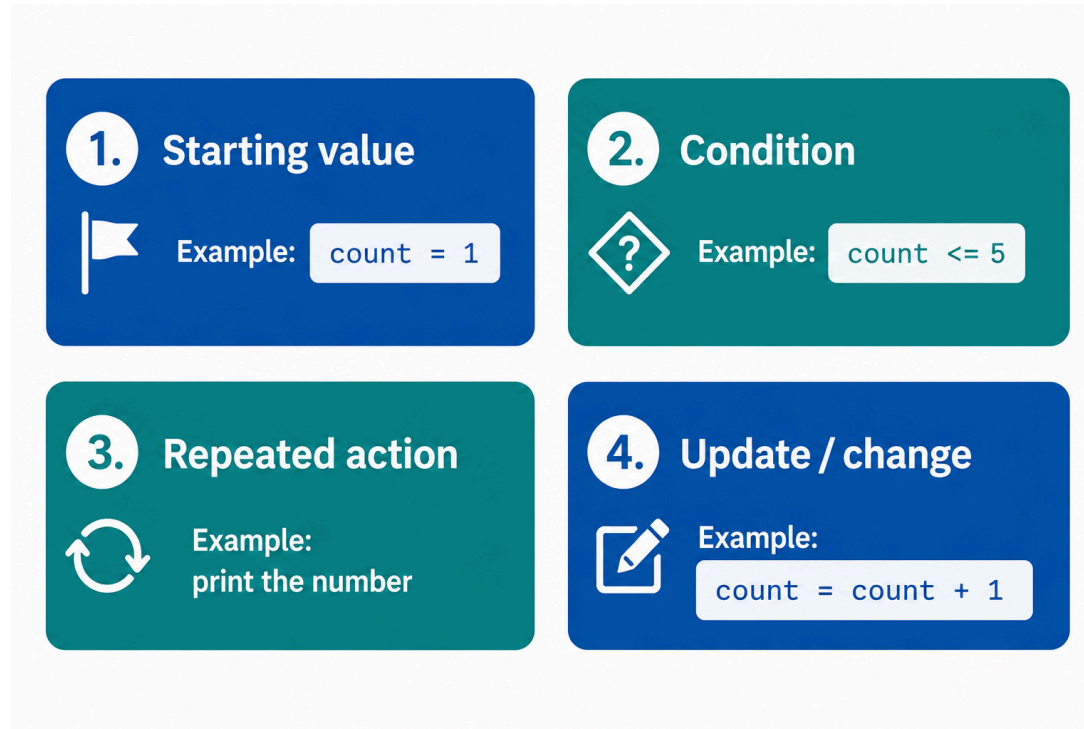
Iteration

- Condition checked repeatedly
- Instructions may run many times
- Stops when condition becomes false

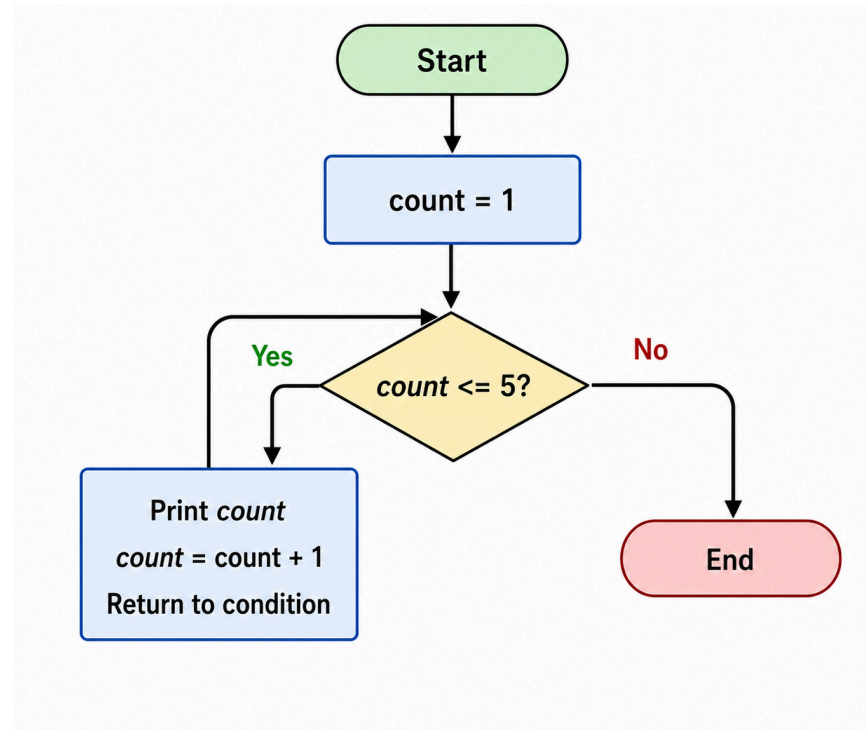
Outline

1. Why do we need loops?
2. Selection versus Iteration
- 3. Anatomy of a loop**
4. Infinite loops
5. Useful loop patterns
6. Activities

3.1 Every loop needs four things



3.2 Counting to five: flowchart



3.3 Trace the loop

count	Condition	Output
1	1 <= 5 is True	1
2	2 <= 5 is True	2
3	3 <= 5 is True	3
4	4 <= 5 is True	4
5	5 <= 5 is True	5
6	6 <= 5 is False	Stop

3.4 Why does the loop stop?

The loop stops because the value changes.

count eventually becomes 6.

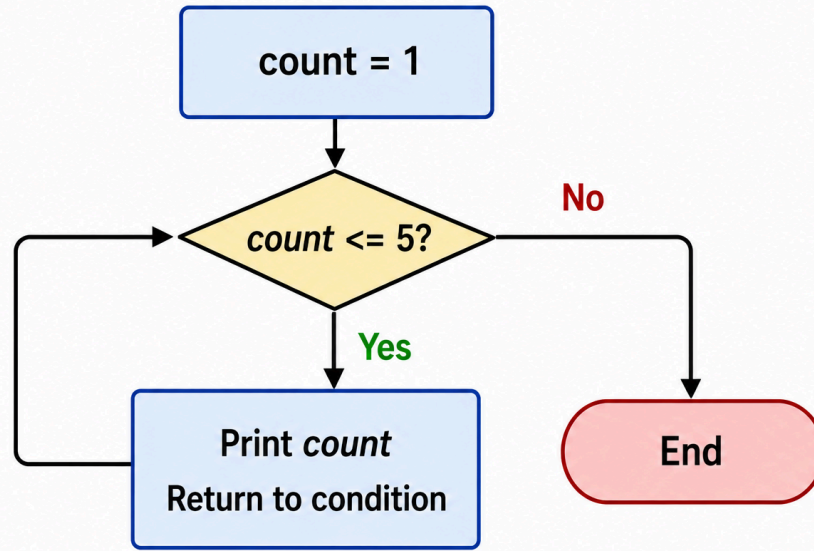
The condition `count <= 5` becomes false.

Important: a loop normally needs something inside it that moves it closer to stopping.

Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
- 4. Infinite loops**
5. Useful loop patterns
6. Activities

4.1 What if we forget the update?



count never changes, so the loop never ends.

4.2 Infinite loop in code

```
count = 1  
  
while count <= 5:  
    print(count)
```

Problem: `count += 1` is missing.

4.3 Corrected code

```
count = 1

while count <= 5:
    print(count)
    count += 1
```

Outline

1. Why do we need loops?

2. Selection versus Iteration

3. Anatomy of a loop

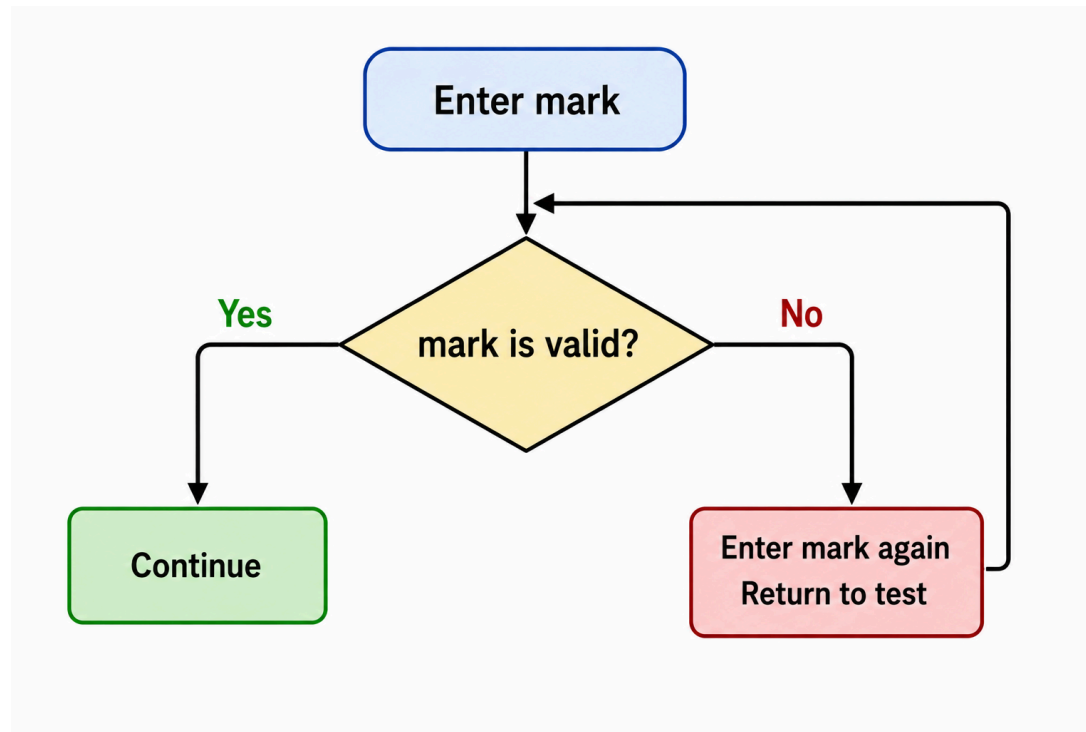
4. Infinite loops

5. Useful loop patterns

6. Activities

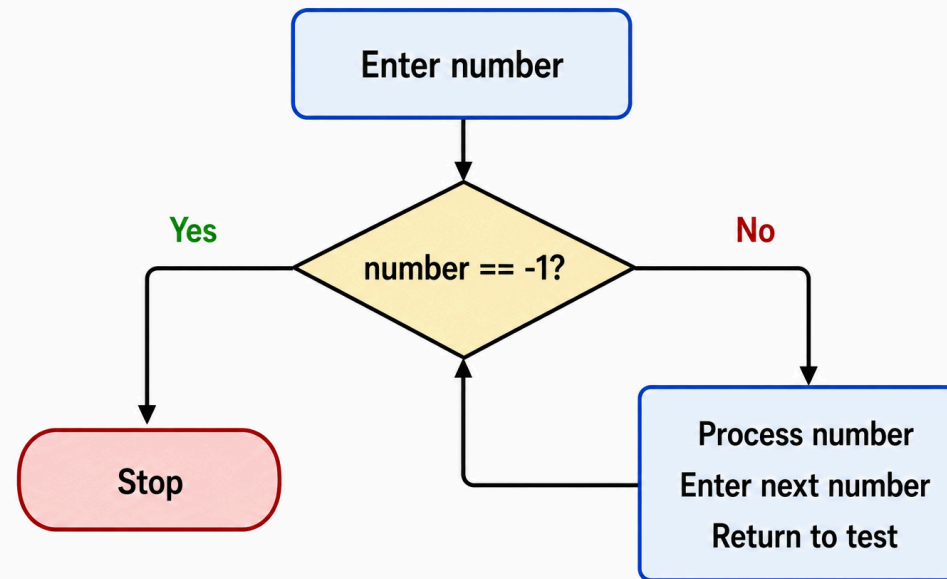
5.1 Validation loop

Use a loop when the user must enter valid data.



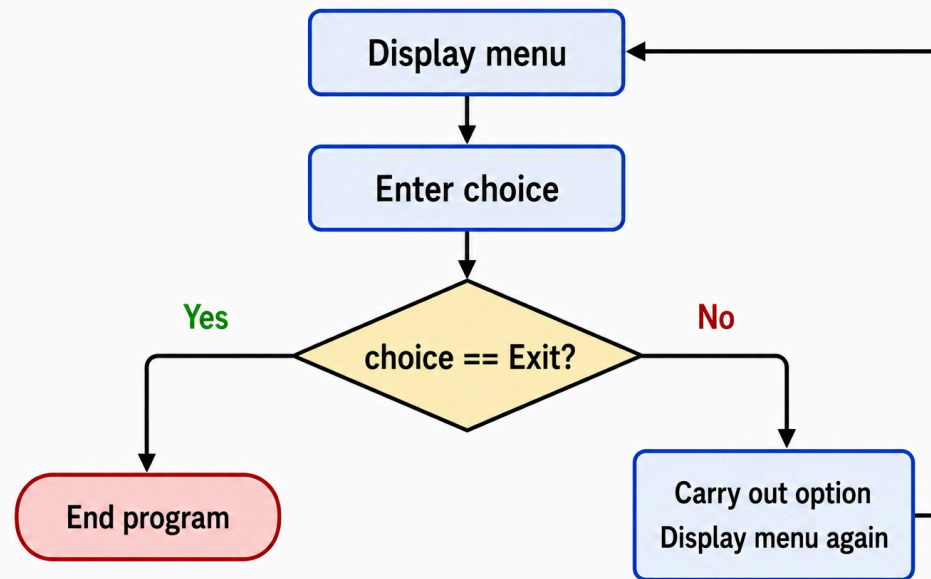
5.2 Sentinel loop

A **sentinel** is a special value that means “stop”.



5.3 Menu loop

Many programs repeat a menu until the user chooses **Exit**.



Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
4. Infinite loops
5. Useful loop patterns
- 6. Activities**

6.1 Activity: find the four parts

For each flowchart, identify:

1. Starting value
2. Condition
3. Repeated action
4. Update or stopping action

This prepares you to write the Python version in the next lecture.

6.2 Predict the output

Flowchart summary:

- Set count = 1
- While count <= 3
- Print "SETU"
- Add 1 to count

Question: what is printed?

6.3 Summary

- Iteration means repetition.
- Flowcharts help us plan loops before writing code.
- Selection asks once; iteration asks repeatedly.
- Every loop needs a condition and a way to stop.
- Missing updates can cause infinite loops.

Next: implementing these ideas in Python using `while` loops.

**Thanks for Watching - Any
questions?**

